Worksheet 1.7 — Mathematical Proof

Mathematics: Analysis & Approaches SL (IB) · Unit 1: Number and Algebra

The Big Ideas

A proof must hold for every case a claim covers; a single counterexample is enough to disprove a claim.

- **Direct proof**: start from known facts/definitions and reason forward to the conclusion.
- **Proof by exhaustion**: check every case, when there are only finitely many.
- **Disproof by counterexample**: one case where the claim fails is enough.

Worked Examples

Example 1 — Direct proof

Prove that the sum of two consecutive even integers is always divisible by 4... or is it? Check first, then prove or disprove.

Solution. **Step 1 (test):** $2+4=6$ (not divisible by 4) — the claim is false as stated.

Step 2: Let the two consecutive even integers be $2n$ and $2n+2$. Their sum is $4n+2=2(2n+1)$, which is divisible by 2 but not generally by 4 (since $2n+1$ is odd).

Conclusion: the original claim is false; the sum is always divisible by 2, but not always by 4.

Example 2 — Disproof by counterexample

Disprove: "For every integer n, n^2+n+1 is odd."

Solution. Test $n=1$: $1+1+1=3$ (odd). Test $n=2$: $4+2+1=7$ (odd). Test general parity: if n is even, n^2 even, n even, so n^2+n+1 is even+even+odd=odd. If n is odd, n^2 odd, n odd, so odd+odd+odd=odd.

Conclusion: the claim actually holds for every integer — checking both parities (a proof by exhaustion over the two cases even and odd) proves it rather than disproving it.

Example 3 — Proof by exhaustion

Prove that n^2-n is even for every integer n from 1 to 4, then generalize.

Solution. **Step 1 (exhaustion for $n=1..4$):** $n=1:0$; $n=2:2$; $n=3:6$; $n=4:12$ — all even.

Step 2 (general case): $n^2-n=n(n-1)$, the product of two consecutive integers. One of any two consecutive integers is always even, so their product is always even.

Conclusion: n^2-n is even for every integer n.

Example 4 — Direct proof — algebraic identity

Prove that the difference of squares of two consecutive odd integers is always a multiple of 8.

Solution. **Step 1:** Let the odd integers be $2n+1$ and $2n+3$.

Step 2: $(2n+3)^2-(2n+1)^2=[(2n+3)-(2n+1)][(2n+3)+(2n+1)]=(2)(4n+4)=8n+8=8(n+1)$.

Conclusion: the difference is $8(n+1)$, a multiple of 8 for every integer n.

Example 5 — Disproof needing a less obvious counterexample

Disprove: "For every positive integer n, $2^n > n^2$."

Solution. Test $n = 1$: $2 > 1$ true. Test $n = 2$: $4 = 4$ false (not strictly greater!).

Conclusion: $n = 2$ is a counterexample — $2^2 = 4^2$... more precisely $2^2 = 4 = 2^2$, so $2^n > n^2$ fails at $n = 2$ (equality, not strict inequality).

Example 6 — Proof by exhaustion over a small finite set

Prove that $n^2 + 1$ is never divisible by 4, for $n = 0, 1, 2, 3$ (representing all four residues mod 4).

Solution. $n = 0 : 1$; $n = 1 : 2$; $n = 2 : 5$; $n = 3 : 10$ — remainders mod 4 are $1, 2, 1, 2$, never 0.

Conclusion: since every integer has one of these four remainders when squared and incremented follows the same pattern (by periodicity), $n^2 + 1$ is never divisible by 4.

Example 7 — Direct proof with a clever substitution

Prove that $(n + 1)^2 - n^2$ is always an odd number, for any integer n.

Solution. Expand: $(n + 1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$.

Conclusion: $2n + 1$ is odd for every integer n (one more than an even number).

Example 8 — Disproof involving a subtle algebra claim

Disprove: "For all real a, b, $\sqrt{a^2 + b^2} = a + b$."

Solution. Test $a = 3, b = 4$: LHS $= \sqrt{9 + 16} = \sqrt{25} = 5$; RHS $= 3 + 4 = 7$.

Conclusion: $5 \neq 7$, so the claim is false.

Example 9 — Proof combining exhaustion and direct reasoning

Prove that the remainder when a perfect square is divided by 3 is never 2.

Solution. **Step 1:** Any integer n has remainder $0, 1$, or 2 mod 3.

Step 2: $n = 3k$: $n^2 = 9k^2$, remainder 0. $n = 3k + 1$: $n^2 = 9k^2 + 6k + 1$, remainder 1. $n = 3k + 2$: $n^2 = 9k^2 + 12k + 4$, remainder 1 (since $4 = 3 + 1$).

Conclusion: every case gives remainder 0 or 1, never 2.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Prove that the sum of any three consecutive integers is always divisible by 3.

Question 2

Disprove: "$n^2 + n + 11$ is prime for every positive integer n." (Hint: try $n = 10$.)

Question 3

Prove that the square of any even number is divisible by 4.

Question 4

Prove by exhaustion that every integer from 10 to 14 is either prime or a product of primes.

Question 5

Disprove: "For every real number x, $x^2 > x$."

Question 6

Prove that the product of any two odd integers is odd.

Question 7

Disprove: "For every positive integer n, $3^n > 2^{n+1}$."

Question 8

Prove that $(n+2)^2 - n^2$ is always divisible by 4, for any integer n.

Question 9

Prove by exhaustion that the remainder of a perfect square divided by 4 is never 3 (test $n = 0, 1, 2, 3 \mod 4$).

Question 10

Disprove: "For all real a, b: $(a+b)^2 = a^2 + b^2$."

Question 11 — Challenge

Prove that $n^3 - n$ is always divisible by 6, for every integer n (hint: factor as $n(n-1)(n+1)$ and think about consecutive integers).

Question 12 — Challenge

A student claims "if $a^2 = b^2$ then $a = b$." Determine whether this is true, and prove or disprove it.

Question 13 — Challenge

Prove that the sum of the squares of any two consecutive integers is always odd.

Question Recap & Answer Key

Question Recap

1. Prove that the sum of any three consecutive integers is always divisible by 3.
2. Disprove: "$n^2 + n + 11$ is prime for every positive integer n." (Hint: try $n = 10$.)
3. Prove that the square of any even number is divisible by 4.
4. Prove by exhaustion that every integer from 10 to 14 is either prime or a product of primes.
5. Disprove: "For every real number x, $x^2 > x$."
6. Prove that the product of any two odd integers is odd.
7. Disprove: "For every positive integer n, $3^n > 2^{n+1}$."
8. Prove that $(n+2)^2 - n^2$ is always divisible by 4, for any integer n.
9. Prove by exhaustion that the remainder of a perfect square divided by 4 is never 3 (test $n = 0, 1, 2, 3 \bmod 4$).
10. Disprove: "For all real a, b: $(a+b)^2 = a^2 + b^2$."
11. Prove that $n^3 - n$ is always divisible by 6, for every integer n (hint: factor as $n(n-1)(n+1)$ and think about consecutive integers).
12. A student claims "if $a^2 = b^2$ then $a = b$." Determine whether this is true, and prove or disprove it.
13. Prove that the sum of the squares of any two consecutive integers is always odd.

✓ Answer Key

1. $n + (n+1) + (n+2) = 3n + 3 = 3(n+1)$, a multiple of 3
2. $n = 10$: $100 + 10 + 11 = 121 = 11^2$, not prime — disproved
3. $(2k)^2 = 4k^2$, divisible by 4
4. $10 = 2 \cdot 5$, 11 prime, $12 = 2^2 \cdot 3$, 13 prime, $14 = 2 \cdot 7$ — every case checks out
5. $x = 0.5$: $0.25 < 0.5$ — disproved
6. $(2m+1)(2n+1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$, odd
7. $n = 1$: $3^1 = 3$, $2^2 = 4$; $3 < 4$, so the inequality fails at $n = 1$ — disproved
8. $(n+2)^2 - n^2 = 4n + 4 = 4(n+1)$, a multiple of 4
9. $n = 0 : 0$; $n = 1 : 1$; $n = 2 : 4 \to 0$; $n = 3 : 9 \to 1$ — remainders are $0, 1, 0, 1$, never 3
10. $a = 1, b = 1$: LHS $= 4$, RHS $= 1 + 1 = 2$; $4 \neq 2$ — disproved
11. Among $n-1, n, n+1$ (three consecutive integers) one is divisible by 3 and at least one is even, so their product is divisible by both 2 and 3, hence by 6
12. False: $a = 2, b = -2$ gives $a^2 = b^2 = 4$ but $a \neq b$ — disproved by counterexample
13. $n^2 + (n+1)^2 = 2n^2 + 2n + 1 = 2(n^2 + n) + 1$, one more than an even number, so always odd